```
#Not able to upload the output resultsb because of file size issue
# Import necessary libraries
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.feature_selection import SelectKBest, f_regression, mutual_info_regression
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the dataset
from google.colab import files
files.upload()
```

Choose Files dataset1.csv

**dataset1.csv**(text/csv) - 206317 bytes, last modified: 3/14/2026 - 100% done

```
Saving dataset1.csv to dataset1.csv
{'dataset1.csv':
b'Date,DayOfWeek,Month,Year,Is_Weekend,Is_Holiday,Product_ID,Category,Base_Price,Discount_Percentage,Current_Price,St
```

```
import pandas as pd

df = pd.read_csv("dataset1.csv")
df.head()
```

| | Date | DayOfWeek | Month | Year | Is_Weekend | Is_Holiday | Product_ID | Category | Base_Price | Discount_Percentage | ... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 4/13/2023 | 3 | 4 | 2023 | 0 | 0 | 1002 | Grocery | 410.09 | 0.01 | ... |
| **1** | 12/15/2023 | 4 | 12 | 2023 | 0 | 0 | 1014 | Home | 89.98 | 0.30 | ... |
| **2** | 9/28/2023 | 3 | 9 | 2023 | 0 | 0 | 1015 | Electronics | 404.88 | 0.14 | ... |
| **3** | 4/17/2023 | 0 | 4 | 2023 | 0 | 0 | 1012 | Grocery | 347.97 | 0.43 | ... |
| **4** | 3/13/2023 | 0 | 3 | 2023 | 0 | 0 | 1016 | Grocery | 280.16 | 0.14 | ... |

5 rows × 22 columns

```
df.shape
```

```
(2000, 22)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 22 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Date                    2000 non-null   object
 1   DayOfWeek               2000 non-null   int64
 2   Month                   2000 non-null   int64
 3   Year                    2000 non-null   int64
 4   Is_Weekend              2000 non-null   int64
 5   Is_Holiday              2000 non-null   int64
 6   Product_ID              2000 non-null   int64
 7   Category                2000 non-null   object
 8   Base_Price              2000 non-null   float64
 9   Discount_Percentage     2000 non-null   float64
 10  Current_Price           2000 non-null   float64
 11  Store_ID                2000 non-null   int64
 12  Store_Type              2000 non-null   object
 13  Competitor_Price        2000 non-null   float64
 14  Footfall_Index          2000 non-null   float64
 15  Avg_Temperature         2000 non-null   float64
 16  Rainfall_mm             2000 non-null   float64
 17  Social_Media_Sentiment  2000 non-null   float64
 18  Lead_Time_Days          2000 non-null   int64
 19  Safety_Stock_Level      2000 non-null   int64
 20  Stock_On_Hand           2000 non-null   int64
 21  Daily_Units_Sold        2000 non-null   int64
dtypes: float64(8), int64(11), object(3)
memory usage: 343.9+ KB
```

```
df.describe()
```

| | DayOfWeek | Month | Year | Is_Weekend | Is_Holiday | Product_ID | Base_Price | Discount_Percentage | Current_Price |
|---|---|---|---|---|---|---|---|---|---|
| **count** | 2000.000000 | 2000.000000 | 2000.0 | 2000.000000 | 2000.000000 | 2000.000000 | 2000.00000 | 2000.000000 | 2000.000000 |
| **mean** | 3.029500 | 6.538500 | 2023.0 | 0.284500 | 0.093000 | 1009.886500 | 246.89615 | 0.248020 | 185.692580 |
| **std** | 1.995651 | 3.467208 | 0.0 | 0.451289 | 0.290505 | 5.721012 | 141.81454 | 0.141437 | 114.283594 |
| **min** | 0.000000 | 1.000000 | 2023.0 | 0.000000 | 0.000000 | 1000.000000 | 5.12000 | 0.000000 | 2.970000 |
| **25%** | 1.000000 | 4.000000 | 2023.0 | 0.000000 | 0.000000 | 1005.000000 | 124.45750 | 0.130000 | 90.245000 |
| **50%** | 3.000000 | 7.000000 | 2023.0 | 0.000000 | 0.000000 | 1010.000000 | 242.50000 | 0.250000 | 173.555000 |
| **75%** | 5.000000 | 10.000000 | 2023.0 | 1.000000 | 0.000000 | 1015.000000 | 367.31750 | 0.370000 | 268.332500 |
| **max** | 6.000000 | 12.000000 | 2023.0 | 1.000000 | 1.000000 | 1019.000000 | 499.61000 | 0.500000 | 486.710000 |

```
df.isnull().sum()
```

| | 0 |
|---|---|
| **Date** | 0 |
| **DayOfWeek** | 0 |
| **Month** | 0 |
| **Year** | 0 |
| **Is_Weekend** | 0 |
| **Is_Holiday** | 0 |
| **Product_ID** | 0 |
| **Category** | 0 |
| **Base_Price** | 0 |
| **Discount_Percentage** | 0 |
| **Current_Price** | 0 |
| **Store_ID** | 0 |
| **Store_Type** | 0 |
| **Competitor_Price** | 0 |
| **Footfall_Index** | 0 |
| **Avg_Temperature** | 0 |
| **Rainfall_mm** | 0 |
| **Social_Media_Sentiment** | 0 |
| **Lead_Time_Days** | 0 |
| **Safety_Stock_Level** | 0 |
| **Stock_On_Hand** | 0 |
| **Daily_Units_Sold** | 0 |

**dtype:** int64

```
import pandas as pd

df = pd.read_csv("dataset1.csv")
print(df.head())
print(df.shape)
print(df.columns)
```

```
         Date  DayOfWeek  Month  Year  Is_Weekend  Is_Holiday  Product_ID  \
0   4/13/2023          3      4  2023           0           0        1002
1  12/15/2023          4     12  2023           0           0        1014
2   9/28/2023          3      9  2023           0           0        1015
3   4/17/2023          0      4  2023           0           0        1012
4   3/13/2023          0      3  2023           0           0        1016

      Category  Base_Price  Discount_Percentage  ...  Store_Type  \
```

```
0      Grocery      410.09                 0.01  ...       Urban
1         Home       89.98                 0.30  ...       Rural
2  Electronics      404.88                 0.14  ...       Urban
3      Grocery      347.97                 0.43  ...       Urban
4      Grocery      280.16                 0.14  ...       Urban

   Competitor_Price Footfall_Index  Avg_Temperature  Rainfall_mm  \
0            411.58          127.6             29.1          8.5
1             90.37           56.9              1.9          5.3
2            408.07          191.8             14.5          2.8
3            349.20           71.6             24.4          1.3
4            252.64          185.5             26.0          0.4

   Social_Media_Sentiment  Lead_Time_Days  Safety_Stock_Level  Stock_On_Hand  \
0                    0.63               7                 141            471
1                    0.35              11                  59            263
2                    0.77               4                 111            302
3                    0.86              11                  38            389
4                    0.36              14                  46            262

   Daily_Units_Sold
0                83
1               102
2                89
3                98
4               100

[5 rows x 22 columns]
(2000, 22)
Index(['Date', 'DayOfWeek', 'Month', 'Year', 'Is_Weekend', 'Is_Holiday',
       'Product_ID', 'Category', 'Base_Price', 'Discount_Percentage',
       'Current_Price', 'Store_ID', 'Store_Type', 'Competitor_Price',
       'Footfall_Index', 'Avg_Temperature', 'Rainfall_mm',
       'Social_Media_Sentiment', 'Lead_Time_Days', 'Safety_Stock_Level',
       'Stock_On_Hand', 'Daily_Units_Sold'],
      dtype='object')
```

```
# Convert the Date column from string format to datetime format
# This allows time-based analysis such as trends and grouping by date
df['Date'] = pd.to_datetime(df['Date'])
```

```
# Verify that the Date column is now converted to datetime format
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 22 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Date                    2000 non-null   datetime64[ns]
 1   DayOfWeek               2000 non-null   int64
 2   Month                   2000 non-null   int64
 3   Year                    2000 non-null   int64
 4   Is_Weekend              2000 non-null   int64
 5   Is_Holiday              2000 non-null   int64
 6   Product_ID              2000 non-null   int64
 7   Category                2000 non-null   object
 8   Base_Price              2000 non-null   float64
 9   Discount_Percentage     2000 non-null   float64
 10  Current_Price           2000 non-null   float64
 11  Store_ID                2000 non-null   int64
 12  Store_Type              2000 non-null   object
 13  Competitor_Price        2000 non-null   float64
 14  Footfall_Index          2000 non-null   float64
 15  Avg_Temperature         2000 non-null   float64
 16  Rainfall_mm             2000 non-null   float64
 17  Social_Media_Sentiment  2000 non-null   float64
 18  Lead_Time_Days          2000 non-null   int64
 19  Safety_Stock_Level      2000 non-null   int64
 20  Stock_On_Hand           2000 non-null   int64
 21  Daily_Units_Sold        2000 non-null   int64
dtypes: datetime64[ns](1), float64(8), int64(11), object(2)
memory usage: 343.9+ KB
```

```
# Check if any columns contain missing (null) values
# This is an important data cleaning step before performing analysis
df.isnull().sum()
```

| | 0 |
|---|---|
| Date | 0 |
| DayOfWeek | 0 |
| Month | 0 |
| Year | 0 |
| Is_Weekend | 0 |
| Is_Holiday | 0 |
| Product_ID | 0 |
| Category | 0 |
| Base_Price | 0 |
| Discount_Percentage | 0 |
| Current_Price | 0 |
| Store_ID | 0 |
| Store_Type | 0 |
| Competitor_Price | 0 |
| Footfall_Index | 0 |
| Avg_Temperature | 0 |
| Rainfall_mm | 0 |
| Social_Media_Sentiment | 0 |
| Lead_Time_Days | 0 |
| Safety_Stock_Level | 0 |
| Stock_On_Hand | 0 |
| Daily_Units_Sold | 0 |

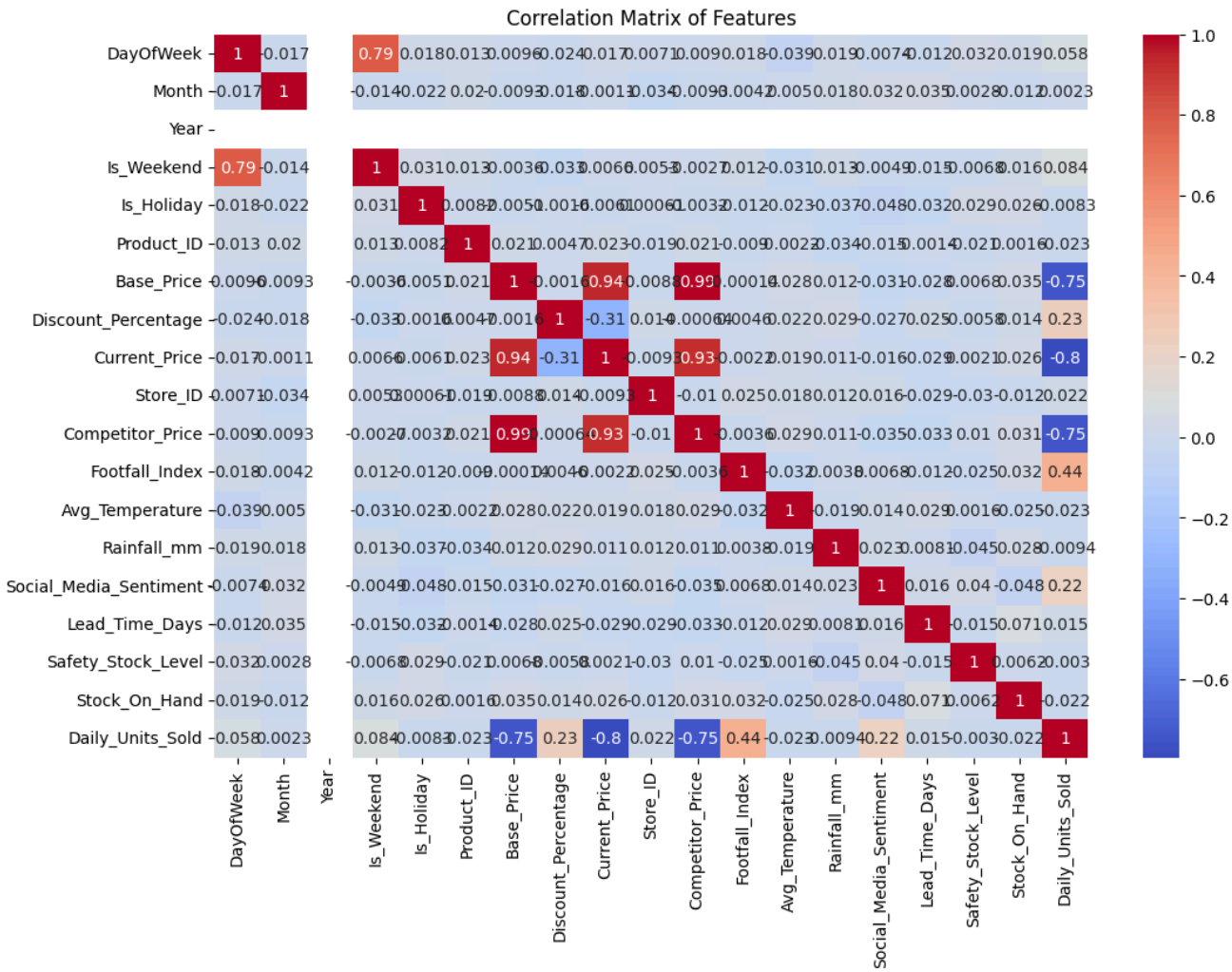
**dtype:** int64

```
# Generate summary statistics for numerical columns
# This includes mean, standard deviation, minimum, maximum, and quartiles
df.describe()
```

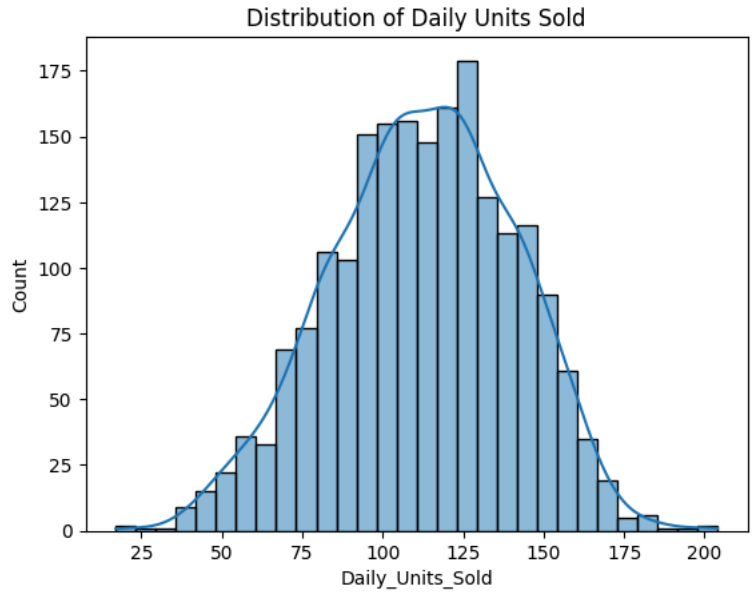
| | Date | DayOfWeek | Month | Year | Is_Weekend | Is_Holiday | Product_ID | Base_Price | Discount_Percenta |
|---|---|---|---|---|---|---|---|---|---|
| count | 2000 | 2000.000000 | 2000.000000 | 2000.0 | 2000.000000 | 2000.000000 | 2000.000000 | 2000.00000 | 2000.0000 |
| mean | 2023-07-02 05:49:12.000000256 | 3.029500 | 6.538500 | 2023.0 | 0.284500 | 0.093000 | 1009.886500 | 246.89615 | 0.2480 |
| min | 2023-01-01 00:00:00 | 0.000000 | 1.000000 | 2023.0 | 0.000000 | 0.000000 | 1000.000000 | 5.12000 | 0.0000 |
| 25% | 2023-04-04 18:00:00 | 1.000000 | 4.000000 | 2023.0 | 0.000000 | 0.000000 | 1005.000000 | 124.45750 | 0.1300 |
| 50% | 2023-07-01 00:00:00 | 3.000000 | 7.000000 | 2023.0 | 0.000000 | 0.000000 | 1010.000000 | 242.50000 | 0.2500 |
| 75% | 2023-10-02 06:00:00 | 5.000000 | 10.000000 | 2023.0 | 1.000000 | 0.000000 | 1015.000000 | 367.31750 | 0.3700 |
| max | 2023-12-31 00:00:00 | 6.000000 | 12.000000 | 2023.0 | 1.000000 | 1.000000 | 1019.000000 | 499.61000 | 0.5000 |
| std | NaN | 1.995651 | 3.467208 | 0.0 | 0.451289 | 0.290505 | 5.721012 | 141.81454 | 0.1414 |

```
# Import visualization libraries for plotting
import seaborn as sns
import matplotlib.pyplot as plt

# Compute correlation matrix for numerical features
corr = df.corr(numeric_only=True)

# Create a heatmap to visualize relationships between variables
plt.figure(figsize=(12,8))
```

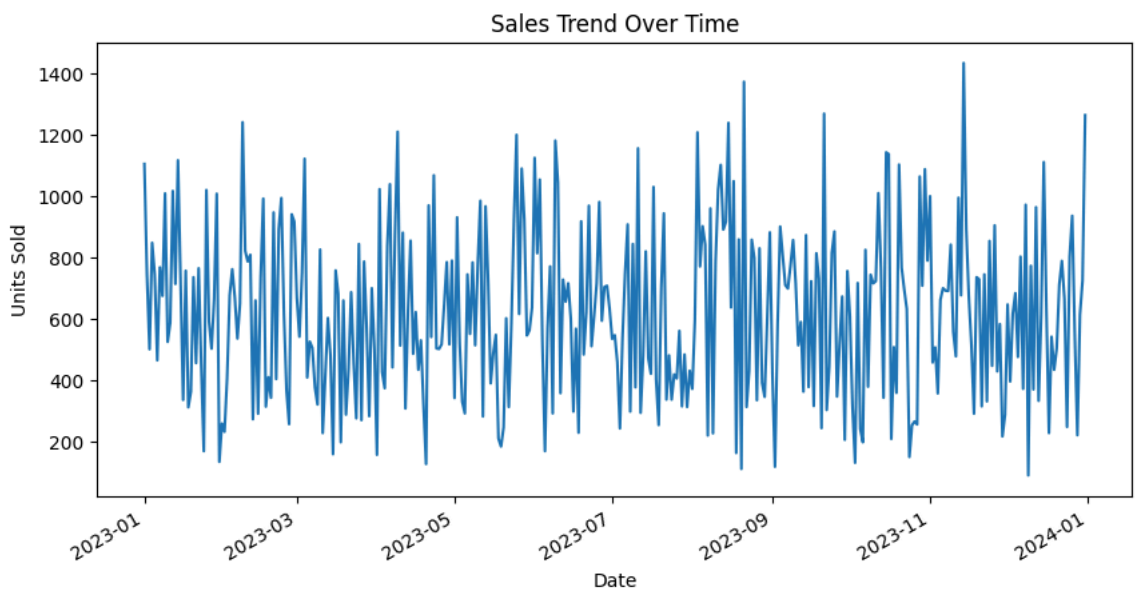
```
plt.figure(figsize=(12,8))
sns.heatmap(corr, annot=True, cmap='coolwarm')

# Add title to the plot
plt.title("Correlation Matrix of Features")

# Display the heatmap
plt.show()
```




```
# Plot the distribution of the target variable
# This helps understand how sales values are spread across the dataset

sns.histplot(df['Daily_Units_Sold'], kde=True)

# Add title to the plot
plt.title("Distribution of Daily Units Sold")

# Display the plot
plt.show()
```




```
# Group sales by date and calculate total units sold per day
sales_trend = df.groupby('Date')['Daily_Units_Sold'].sum()

# Plot the sales trend over time
sales_trend.plot(figsize=(10,5))

# Add labels and title
plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Units Sold")

# Display the plot
plt.show()
```




```
# Import StandardScaler to normalize numerical features
from sklearn.preprocessing import StandardScaler

# Remove columns that should not be used for PCA
# (Date and target variable are excluded)
features = df.drop(columns=['Date', 'Daily_Units_Sold'])

# Convert categorical variables into numeric format
features = pd.get_dummies(features, drop_first=True)

# Initialize the scaler
scaler = StandardScaler()
```

```
# Fit and transform the data
scaled_features = scaler.fit_transform(features)
```

```
# Import PCA from sklearn
from sklearn.decomposition import PCA

# Initialize PCA model
pca = PCA()

# Apply PCA on the scaled dataset
pca_components = pca.fit_transform(scaled_features)
```

```
# Check how much variance each principal component explains
explained_variance = pca.explained_variance_ratio_

# Print first five components
print("Explained variance by first 5 components:")
print(explained_variance[:5])
```

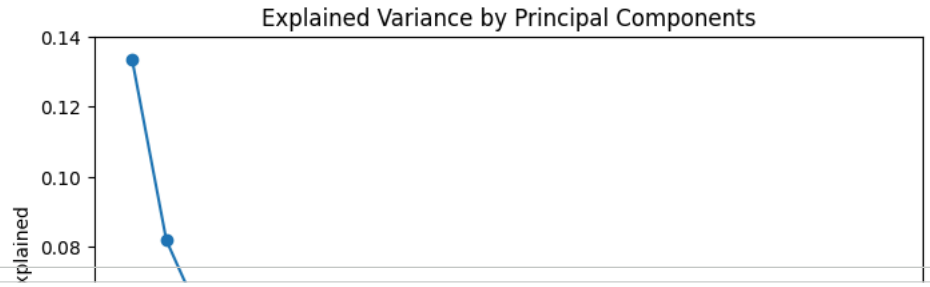
```
Explained variance by first 5 components:
[0.13339154 0.08185988 0.06006487 0.05804271 0.05754784]
```

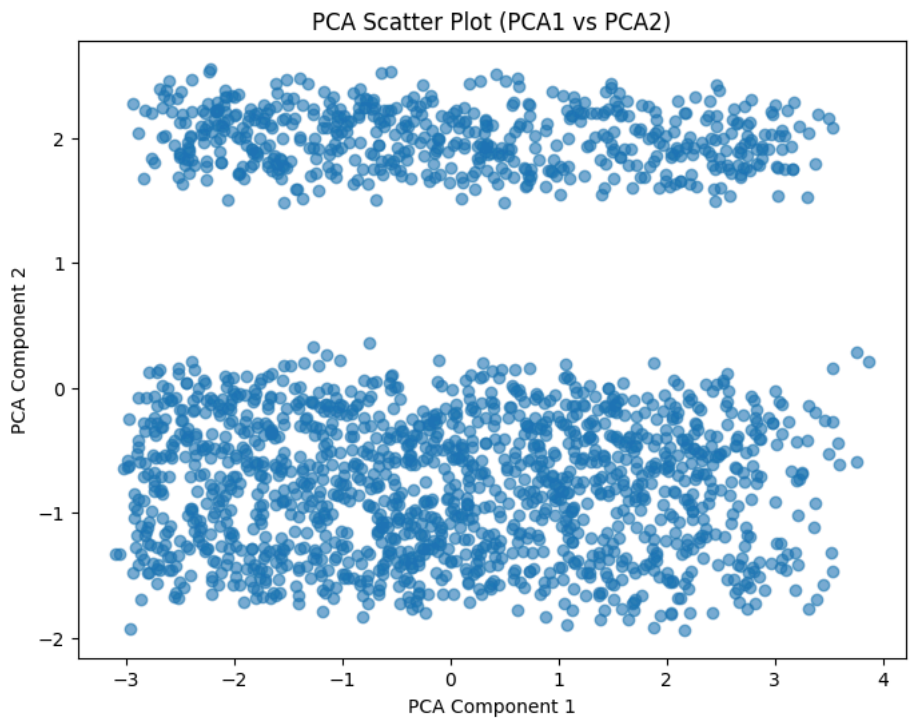
```
# Add the first three principal components as new features in the dataset
df['PCA1'] = pca_components[:,0]
df['PCA2'] = pca_components[:,1]
df['PCA3'] = pca_components[:,2]

# Display updated dataset
df.head()
```

| | Date | DayOfWeek | Month | Year | Is_Weekend | Is_Holiday | Product_ID | Category | Base_Price | Discount_Percentage | ... | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 2023-04-13 | 3 | 4 | 2023 | 0 | 0 | 1002 | Grocery | 410.09 | 0.01 | ... | |
| **1** | 2023-12-15 | 4 | 12 | 2023 | 0 | 0 | 1014 | Home | 89.98 | 0.30 | ... | |
| **2** | 2023-09-28 | 3 | 9 | 2023 | 0 | 0 | 1015 | Electronics | 404.88 | 0.14 | ... | |
| **3** | 2023-04-17 | 0 | 4 | 2023 | 0 | 0 | 1012 | Grocery | 347.97 | 0.43 | ... | |
| **4** | 2023-03-13 | 0 | 3 | 2023 | 0 | 0 | 1016 | Grocery | 280.16 | 0.14 | ... | |

5 rows × 25 columns

```
# Plot explained variance of principal components
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.plot(pca.explained_variance_ratio_, marker='o')

plt.title("Explained Variance by Principal Components")
plt.xlabel("Principal Component")
plt.ylabel("Variance Explained")

plt.show()
```




```
# Visualize the relationship between the first two principal components

plt.figure(figsize=(8,6))

# Create a scatter plot of PCA1 vs PCA2
plt.scatter(df['PCA1'], df['PCA2'], alpha=0.6)

# Add labels and title
plt.xlabel("PCA Component 1")
plt.ylabel("PCA Component 2")
plt.title("PCA Scatter Plot (PCA1 vs PCA2)")

# Display the plot
plt.show()
```